



13.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to teach them how to use coordinate geometry to justify congruence and similarity in polygons.

Unit 13, Lesson 13: Finding Perimeter and Area on the Coordinate Plane



Benchmark

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.



Learning Target

- ☐ I can find the perimeter and area of polygons and composite figures drawn on the coordinate plane.



13.13.1 Warm-Up: The Dancing House

The Dancing House is located in Prague, Czech Republic, and was designed in 1992. The style of this building is known as deconstructivist architecture due to its unusual shape.

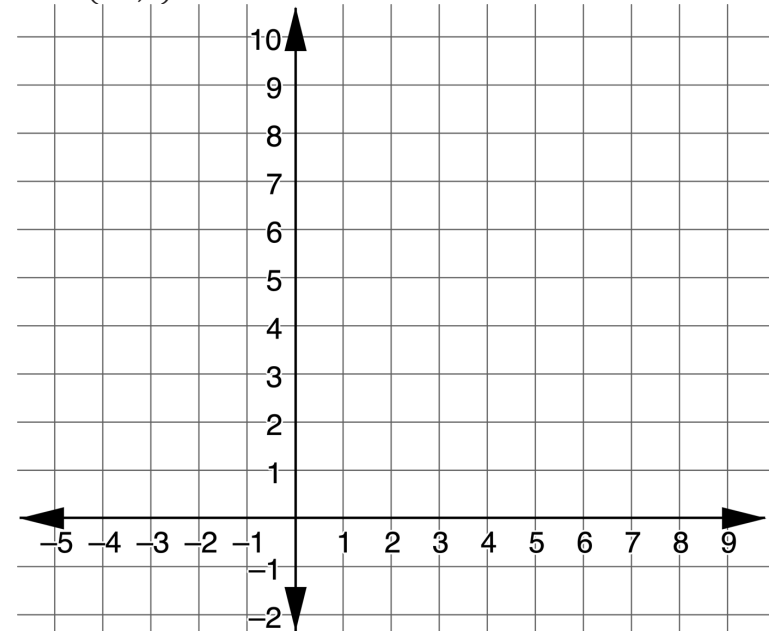


1. What is the first question that comes to your mind?
2. Write a question that a 5th grader could answer using the photo.



13.13.2 Exploration: Perimeter and Area

1. Trapezoid $BGMH$ has vertices at $B(1,9)$, $G(5,9)$, $M(2,5)$, and $H(-4,5)$.



- a. Graph $BGMH$.
- b. Complete the table.

Line Segment	Length
$\overline{BG}$	
$\overline{GM}$	
$\overline{MH}$	
$\overline{HB}$	

c. What is the perimeter of $BGMH$?

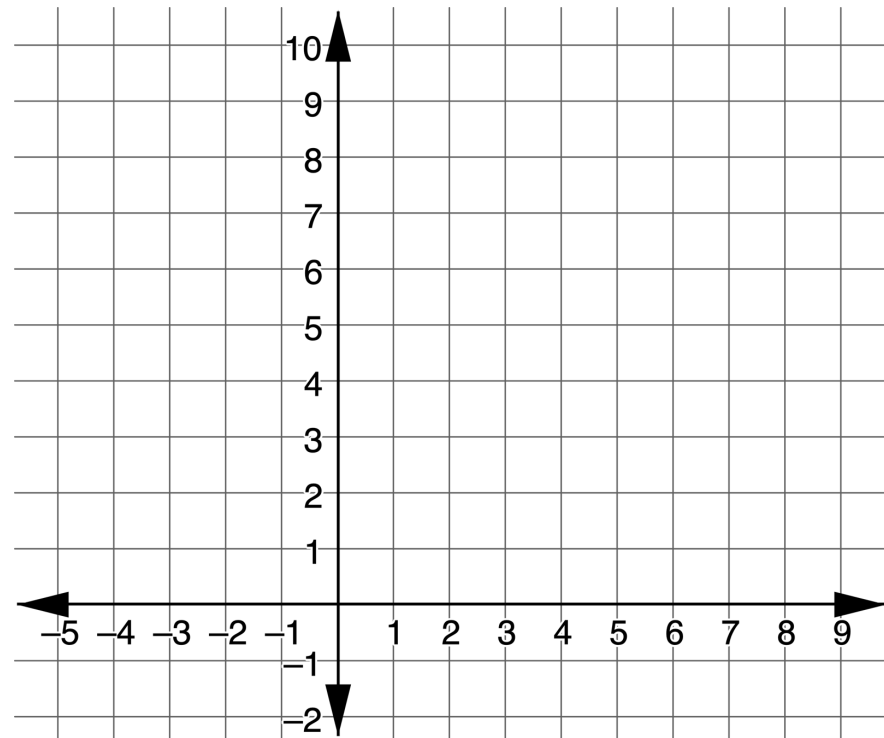
d. Draw a diagonal line from B to M to make $\triangle BMH$ and $\triangle BGM$.

e. Find the area of $\triangle BMH$ and $\triangle BGM$.

$\triangle BMH =$ _____ $\triangle BGM =$ _____

f. What is the area of $BGMH$?

2. Parallelogram $GPKM$ has vertices at $G(5,9)$, $P(5,6)$, $K(2,2)$, and $M(2,5)$.



a. Graph $GPKM$.

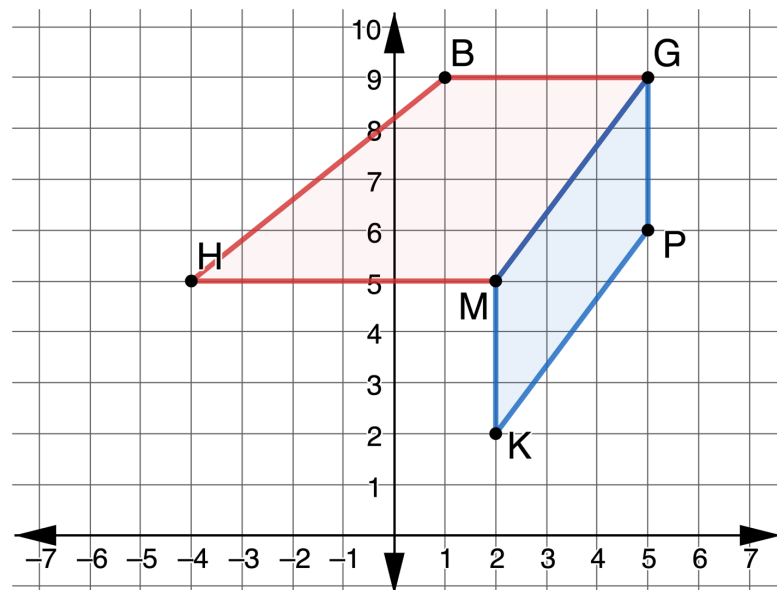
b. Complete the table.

Line Segment	Length
$\overline{GP}$	
$\overline{PK}$	
$\overline{KM}$	
$\overline{MG}$	

c. What is the perimeter of $GPKM$?

d. What is the area of $GPKM$?

3. The figure shown is created by trapezoid $BGMH$ and parallelogram $GPKM$.



a. What do you notice about this figure?



A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

b. Explain why this figure is considered a composite figure.

c. How could you find the perimeter of a composite figure?

d. What is the perimeter of composite figure $BGPCKMH$?

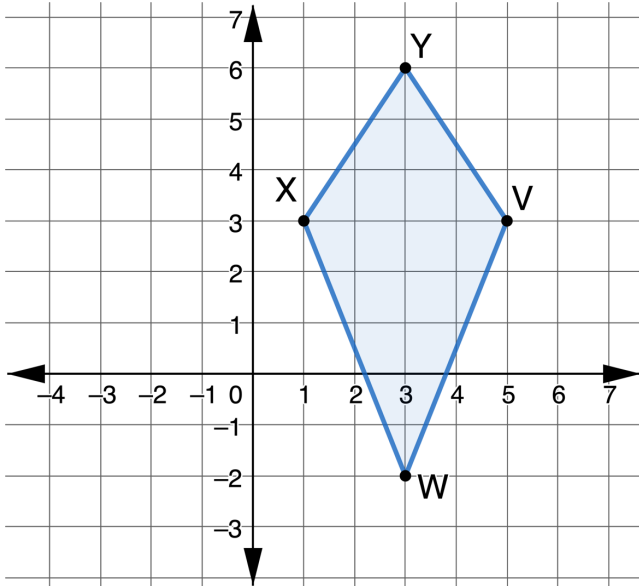
e. Explain why the perimeter of the composite figure is not the sum of the perimeters of the parallelogram and the trapezoid.

f. What is the area of $BGPCKMH$?



13.13.3 Try It: Perimeter and Area

1. Kite $YVWX$ is shown with vertices at $Y(3,6)$, $V(5,3)$, $W(3,-2)$, and $X(1,3)$.



- a. Complete the table.

Line Segment	Length
$\overline{YV}$	
$\overline{VW}$	
$\overline{WX}$	
$\overline{XY}$	

- b. What is the perimeter of $YVWX$?

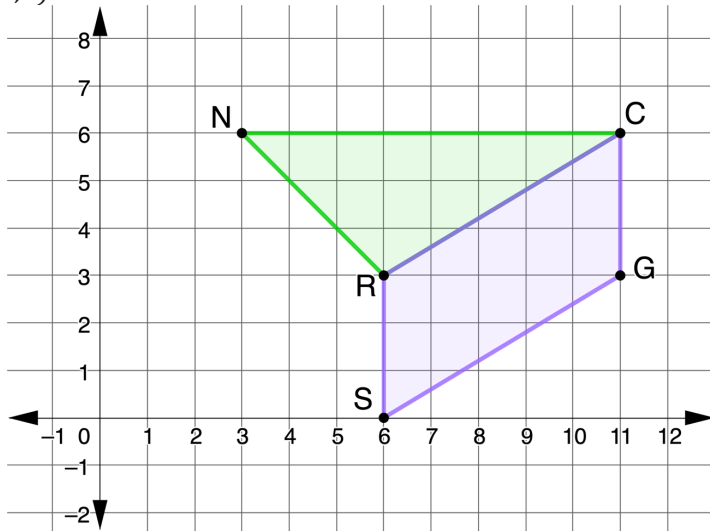
- c. Draw line XV to create $\triangle YVX$ and $\triangle VWX$.

- d. Find the area of each triangle.

$\triangle YVX =$ _____ $\triangle VWX =$ _____

- e. What is the area of $YVWX$?

2. The composite figure shown is created by $\triangle NCR$ and parallelogram $RCGS$. $N(3,6)$, $C(11,6)$, $G(11,3)$, $S(6,0)$, and $R(6,3)$.



- a. Complete the table. Round your answer to the nearest thousandth.

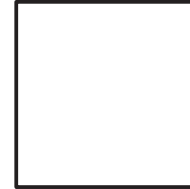
Figure	Perimeter	Area
$\triangle NCR$		
$RCGS$		

- b. What is the perimeter of the composite figure?
- c. What is the area of the composite figure?

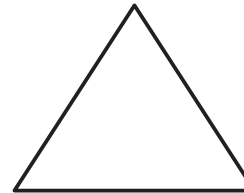


13.13.4 Wrap-Up: Cool Down

1. What is something that is now “squared away” in your mind about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



2. What are three “points” that are important about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



3. What is something that is still “circling” your brain about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?

